Software Requirements Specification

Household Carbon Footprint & Target Tracker

Problem Statement #61 — Sustainability & Green Tech

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the Household Carbon Footprint & Target Tracker, a green-living analytics tool that lets households log monthly utility consumption and track progress toward self-defined carbon-reduction targets. It is intended for the development team, instructors evaluating the project, and any future maintainers of the system.

1.2 Scope

The system allows a Resident to log monthly electricity (kWh), gas usage, and commute distance (km), converts these into CO2 equivalent (CO2e) emissions, and tracks progress toward a reduction target the Resident sets. A Sustainability Coach can view a Resident's data and submit written recommendations. The system does not cover billing, hardware metering integration, or multi-tenant organization management — it is scoped to the household/coach relationship described in the assigned problem statement.

1.3 Definitions, Acronyms and Abbreviations

- CO2e — Carbon dioxide equivalent, a standardized unit for comparing emissions from different greenhouse gases.
- FR — Functional Requirement.
- NFR — Non-Functional Requirement.
- EPA — (U.S.) Environmental Protection Agency, source of the emissions conversion factors used in this system.
- Resident — The primary end-user who logs household consumption data.
- Sustainability Coach — A secondary, advisory user who reviews Resident data and provides guidance.

2. Overall Description

2.1 Product Perspective

The Tracker is a standalone, self-contained web/mobile analytics application. It is not a component of a larger product suite. It depends on externally maintained EPA conversion-factor data to compute emissions accurately.

2.2 User Classes and Characteristics

- Resident — A household member with no specialized technical knowledge; interacts with the system monthly to log data and periodically to check progress.
- Sustainability Coach — A more engaged or trained user who reviews multiple households' data and provides written guidance; assumed to have a basic understanding of emissions reduction strategies.

2.3 Operating Environment

The system is expected to run as a responsive web application (and/or companion mobile view) accessible via standard modern browsers, backed by a database that persists consumption logs, targets, and recommendations per household.

2.4 Assumptions and Dependencies

- EPA conversion factors are available and periodically updated by an external, trusted source.
- Each Resident belongs to exactly one household; each household has at most one assigned Sustainability Coach for the scope of this project.
- Network connectivity is available when a Resident logs data or a Coach reviews it; offline logging is out of scope.

3. System Features (Functional Requirements)

The following functional requirements were elicited from the assigned problem statement and are detailed further, with acceptance criteria and rationale, in the accompanying Requirements Table and Requirements Traceability Matrix (RTM).

ID	Description	Priority
FR-001	Ingest household monthly electricity (kWh) and transport (km) logs to compute total CO2e emissions.	High
FR-002	Allow the Resident to set a monthly or annual carbon-reduction target.	High
FR-003	Calculate and display percentage progress toward the active target after each new log entry.	High
FR-004	Notify the Resident when a progress milestone (25/50/75/100%) is reached.	Medium
FR-005	Allow a Sustainability Coach to view household data and submit recommendations.	Medium

4. External Interface Requirements

4.1 User Interfaces

A web-based (and/or mobile-responsive) dashboard presenting: a monthly data-entry form (electricity, gas, commute), a progress view against the active target, a historical comparison chart, and a Coach-facing household review screen.

4.2 Software Interfaces

The system integrates with an external EPA conversion-factor reference (data feed or periodically updated internal table) to compute CO2e from raw consumption figures.

4.3 Communications Interfaces

Standard HTTPS communication between the client (browser/mobile) and the backend services; no other external communication protocols are required for the current scope.

5. Non-Functional Requirements

ID	Description	Priority
NFR-001	The annual carbon-reduction milestone chart must render interactive historical comparisons in under 200ms under simulated peak load.	High
NFR-002	Household data shall be encrypted at rest and in transit, accessible only to the owning Resident and their assigned Coach.	High

6. Appendices

Appendix A — UML Use-Case Diagram: see Household_Carbon_Tracker_UseCase_Diagram.pdf (Lab 1 deliverable).

Appendix B — UML Component Diagram: see Household_Carbon_Tracker_Component_Diagram.pdf (Lab 3 deliverable).

Appendix C — Requirements Traceability Matrix: see Household_Carbon_Tracker_RTM.pdf.